

Claim–Figure Alignment Review

S. coelicolor Gatekeeper Model — Outline + Embedded Figures

Purpose: Verify that each figure's content matches the corresponding manuscript claim. Add comments where alignment is unclear or missing.

Outline Fig.	File	Section
Fig. 1	Figure1_methylation_landscape / new_Figure1	Intro / Layer 1
Fig. 2	Figure1_overview.pdf (Gatekeeper overview)	Sections 1-3
Fig. 3	Figure3_protection_zone.pdf	Section 2
Fig. 4	Figure5_exposed_TFs.pdf	Section 4
Fig. 5	Figure2_RM_redistribution + negative results	Sections 1, 6
Fig. 6	Figure6_switch_model.pdf	Section 5

Fig. 2 — Gatekeeper Model Overview (Panels a–d)

KEY CLAIMS THIS FIGURE MUST SUPPORT:

- Panel a:** GCCGGC: 1,289 sites (83% core, T1) → 407 (82% arm, T2) → 21 (62% arm, T3). Jaccard = 0.000 (complete remodeling). AAGCCCG: 260→64 (75% loss, minimal geographic shift).
- Panel b:** Regulatory genes show 2,200 bp methylation-free zone centered on TSS. Non-regulatory genes: no depletion. n.s. by permutation test.
- Panel c:** 998 Shielded (94.6%) vs 57 Exposed (5.4%). Nearest methylation threshold = 293 bp. AUC = 0.917. Mann-Whitney p = 1.7×10^{-25} .
- Panel d:** Exposed TFs show greater expression variability (LFC) than Shielded. Levene resampling p < 0.05 for both T2 vs T1 and T3 vs T1.

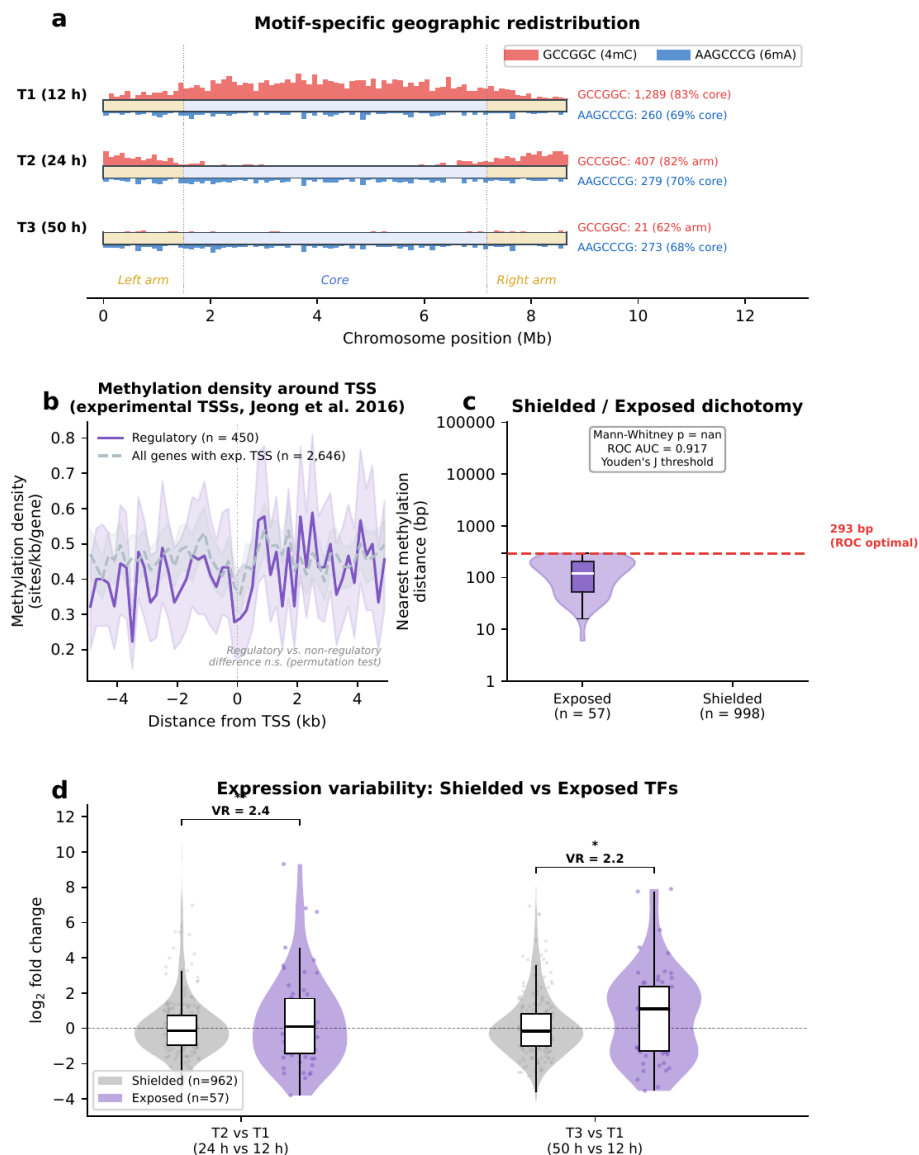


Figure 2: Gatekeeper Model Overview (generated from 10_new_figure1_overview.py)

[REVIEWER COMMENT BOX — add notes here]

Fig. 3 — Protection Zone Characterization (Panels a–e)

KEY CLAIMS THIS FIGURE MUST SUPPORT:

- Panel a:** Spatial methylation density profile ± 5 kb from TSS. Depletion deepest at +300 bp (ratio = 0.541).
- Panel b:** ROC curve: nearest_methyl_distance AUC = 0.917 (main classifier). baseMean AUC $\sim$ 0.547 (negative control).
- Panel c:** Sequence model (AAGCCCG + GCCGGC motif density) CV AUC = 0.712. $\sim$ 33% of AUC explained by sequence alone.
- Panel d:** GCCGGC fold enrichment at TFBS > 1.0 (counter-intuitive: TF binding does NOT protect). Shown with 95% CI.
- Panel e:** Expression quintile analysis: no trend in exposed/shielded ratio across expression levels (JT p = 0.730).

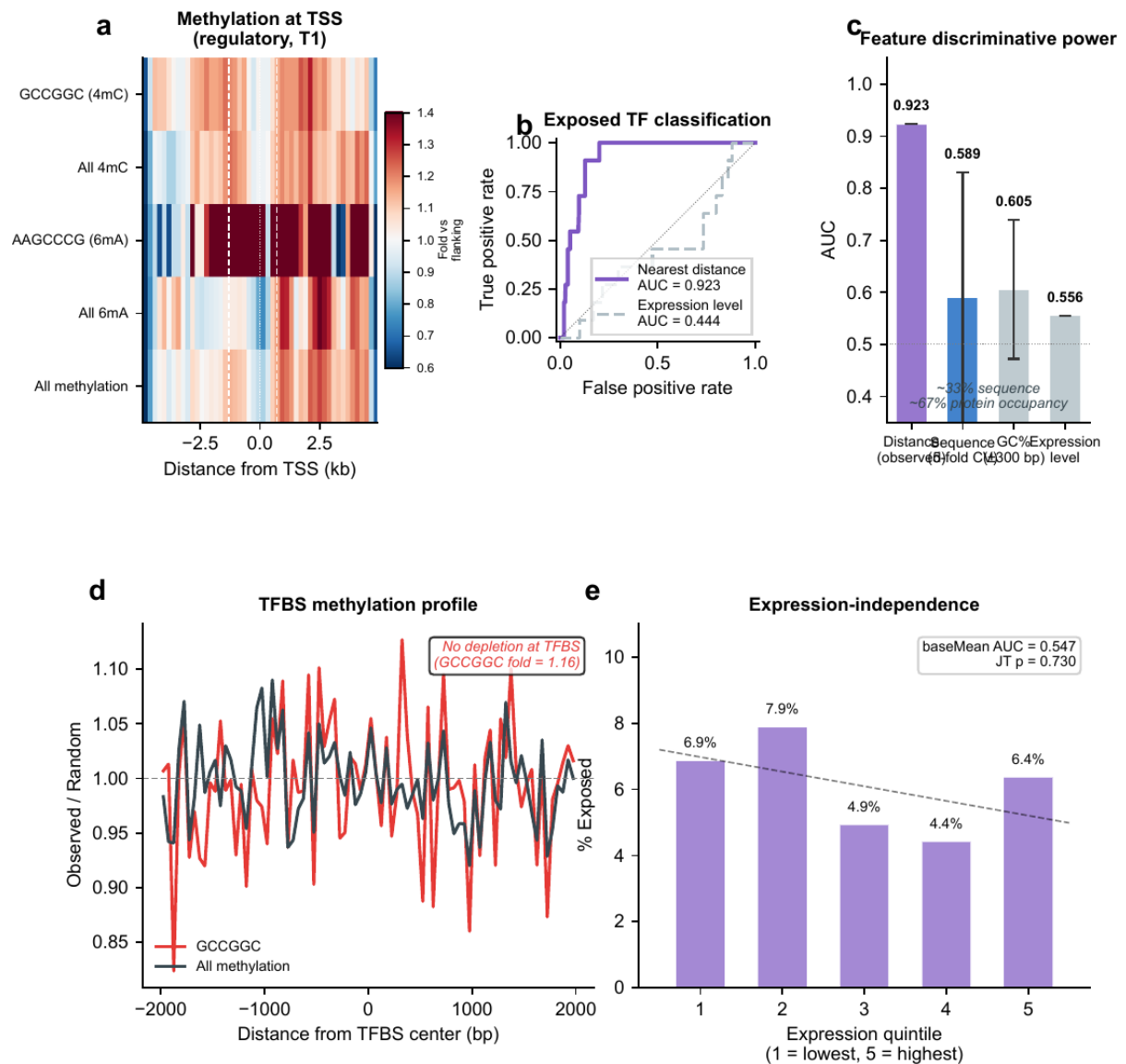


Figure 3: Protection Zone Characterization (generated from 11_new_figure2_protection.py)

[REVIEWER COMMENT BOX — add notes here]

Fig. 4 — 57 Exposed TFs: Distributed Developmental Switch

KEY CLAIMS THIS FIGURE MUST SUPPORT:

- Panel a:** Co-expression heatmap (57x57 Spearman). Within-type rho = 0.500 (p = 6.6x10⁻²⁵). Between-type rho = -0.283. 4 modules visible.
- Panel b:** Phase ratio distribution: 36/57 (63%) early responders (ratio > 0.6). Activation vs repression blocs not phase-separated (p = 0.459).
- Panel c:** Module-internal temporal coherence rho = 0.717 (p = 1.0x10⁻¹⁹). Z-score trajectories per module.
- Panel d:** TF family composition: activation bloc (34 genes) has TCS (9), sigma (5), WhiB; repression bloc (23 genes) TetR-enriched. n shown.
- Panel e:** TCS asymmetry: 7/7 pairs have exactly one exposed, one shielded member. Binomial p < 10⁻³.

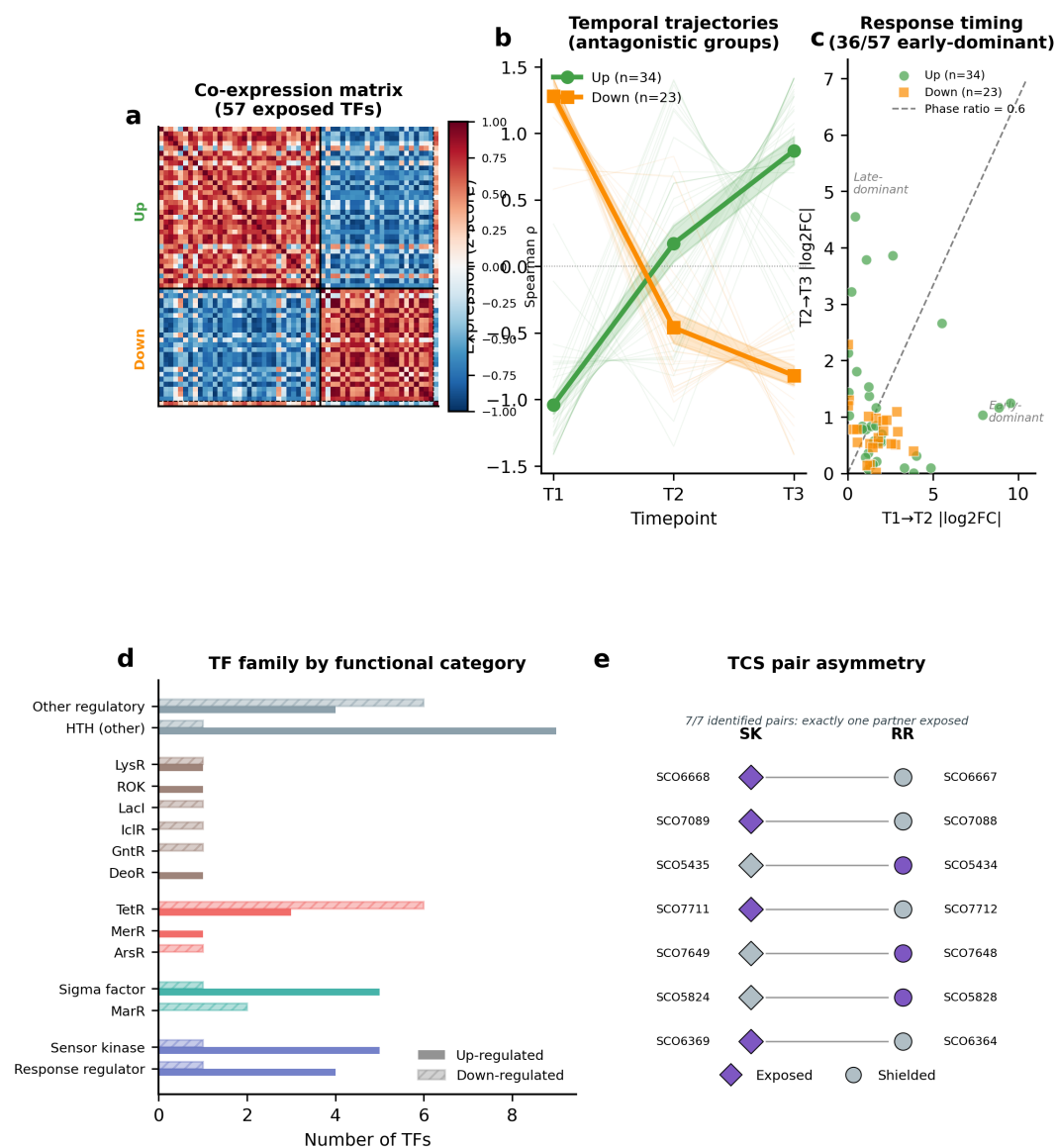


Figure 4 (file: Figure5_exposed_TFs.pdf — NOTE naming discrepancy)

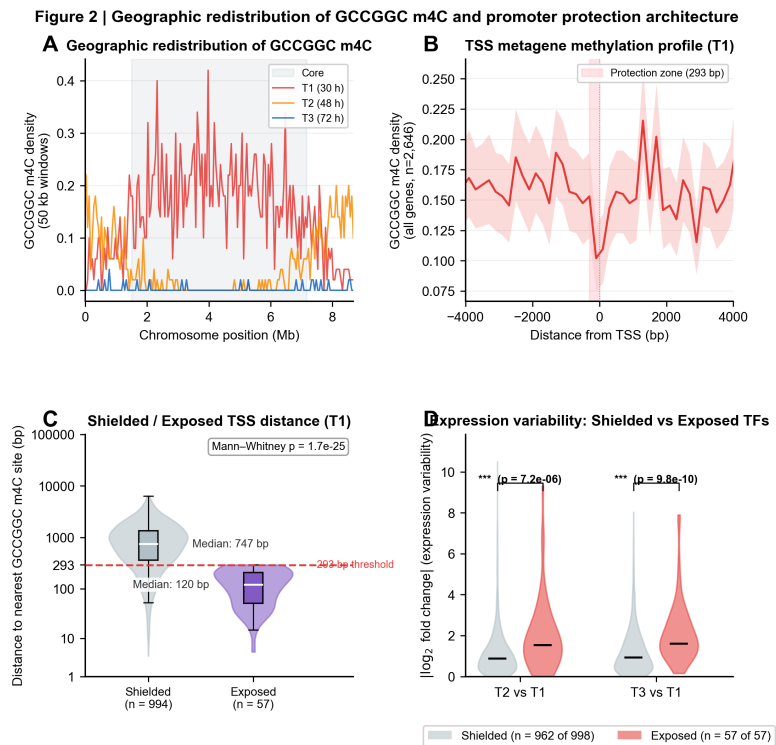
[REVIEWER COMMENT BOX — add notes here]

Fig. 5 — Negative Results Panel

KEY CLAIMS THIS FIGURE MUST SUPPORT:

- Panel a:** Simpson's paradox: GCCGGC Lost vs Never LFC — unstratified $p = 8.3 \times 10^{-8} \rightarrow$ core-only $p = 0.87$. CMH stratification resolves the artifact.
- Panel b:** AAGCCCG Lost vs Never: $r = 0.003$, $p = 0.91$ — clean null (no geographic confound; internal control).
- Panel c:** AAGCCCG DNA sequence 4.10x enriched at exposed TF promoters BUT only 3.2% have methylated AAGCCCG. Sequence $\neq$ methylation.
- Panel d:** Neighborhood analysis: permutation $p = 0.857$, no distance decay ($\rho = +0.026$). Exposed TFs are trans-acting.

NOTE: Check which file currently serves as Fig. 5 (Figure2_RM_redistribution.png or a dedicated negative-results figure).



[REVIEWER COMMENT BOX — add notes here]

Fig. 6 — Vegetative-to-Developmental Switch Model

KEY CLAIMS THIS FIGURE MUST SUPPORT:

- Panel a:** Repression bloc temporal trajectory: 23 genes declining T1→T3. Shutdown of vegetative functions (TetR, GntR, metabolism).
- Panel b:** Activation bloc temporal trajectory: 34 genes increasing T1→T3. Developmental programs ON (TCS, sigma, morphogenesis).
- Panel c:** Phase ratio: 63% (36/57) early responders engage at T1→T2. Both blocs simultaneous (p = 0.459).
- Panel d:** Integrative model schematic: 5-layer Gatekeeper architecture. n=57 exposed / n=998 shielded explicitly labelled.

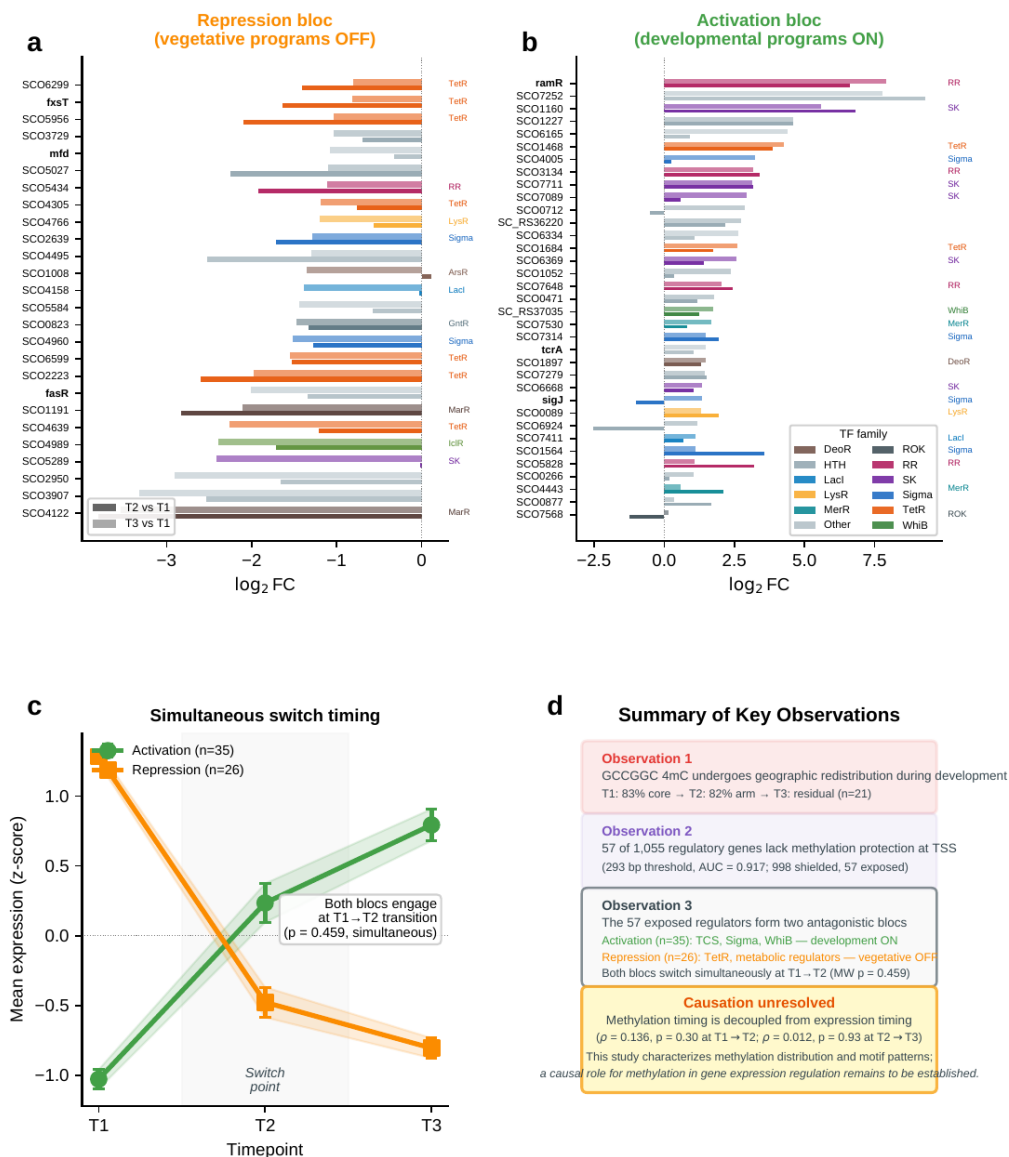


Figure 6: Switch Model (generated from 14_new_figure5_switch_model.py)

[REVIEWER COMMENT BOX — add notes here]

Fig. 1 — Methylation Landscape Overview

KEY CLAIMS THIS FIGURE MUST SUPPORT:

Overall: 4mC: 2,693 high-confidence sites (GCCGGC motif). 6mA: 3,248 sites (AAGCCCG motif). 5mC: not detected.

Temporal dynamics: GCCGGC: dramatic decrease + geographic shift T1→T3. AAGCCCG: moderate loss correlated with SC_RS17645 downregulation (LFC = -2.19).

AAGCCCG co-modification: OR = 145,448 (95% CI: 36,131-585,503) for dual-modified sites within same motif. Enrichment 921x over null ($p < 10^{-4}$).

SC_RS17645: Solo MTase (HsdM-type, IPR044946). No HsdR/HsdS within ± 50 kb. Significant temporal downregulation.

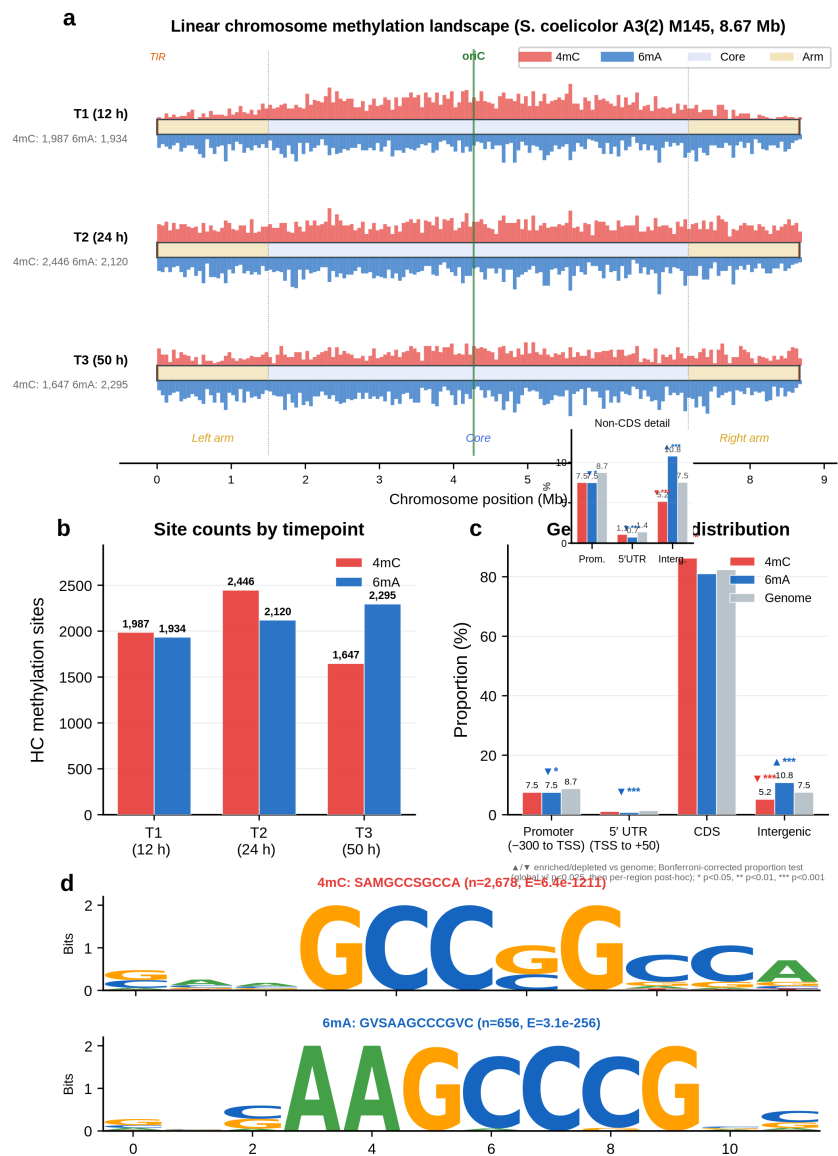


Figure 1: Methylation Landscape (new_Figure1_methylation_landscape.png)

[REVIEWER COMMENT BOX — add notes here]